

Healthcare Analytics Dashboard - Week 5

Power BI project report | Tool: Microsoft Power BI Desktop

1. Project Overview

In this project I turned a healthcare dataset into an interactive dashboard using Power BI. The report brings patient, admission and billing details together, so the key numbers can be explored and compared with filters and charts.

In Week 5 I set up the data model, created the measures and arranged everything neatly on one report page.

The main things I looked at were:

- Number of patients and their demographics
- Admission types and admission counts
- Billing amount and average billing
- Analysis by medical condition
- Gender and age group analysis
- Filtering by date and other fields

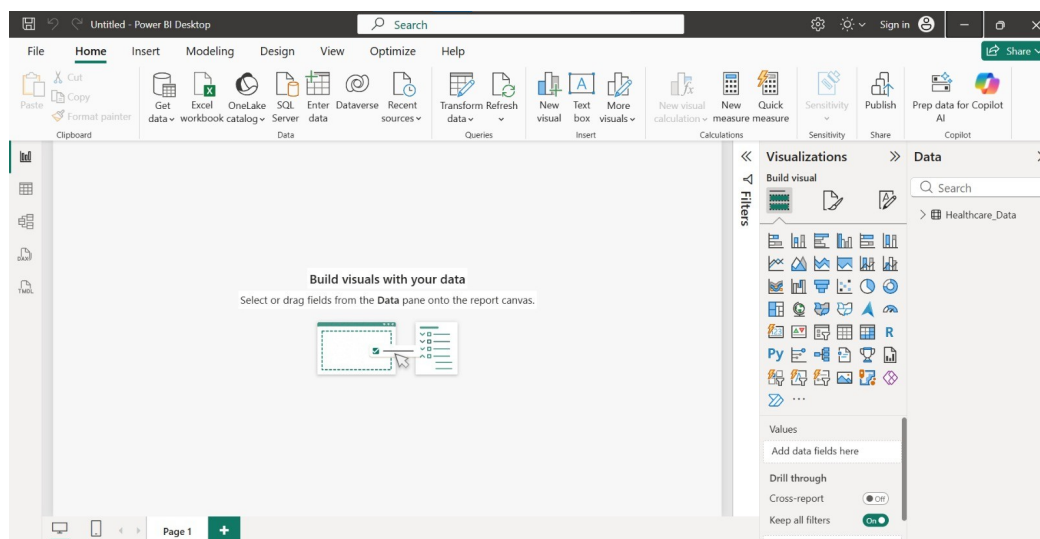


Figure 1: Blank report canvas before building the dashboard

2. Data Model and Relationships

I split the data into separate tables in the model. There is the main Healthcare_Data table, two dimension tables (Dim_Patient and Dim_Admission) and a Billing table. Dim_Patient holds age, age group, gender, blood type, name and patient ID. Dim_Admission has the admission type, doctor, hospital, medical condition, medication, room number and test results. The Billing table has the billing amount, admission and discharge dates and the insurance provider.

Both dimension tables are connected to the Billing table with one-to-many relationships, so a filter on any visual works across all the tables.

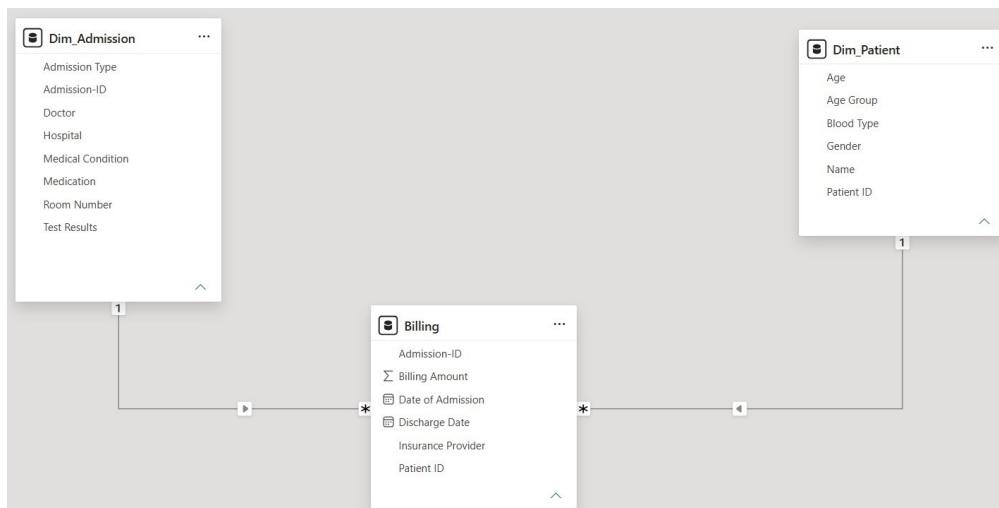


Figure 2: Model view showing the relationships

3. Measures

I created DAX measures so the summary values update whenever a filter is changed. The same measures are used in the cards and in the charts. They are:

- Total Billing - total of the billing amount
- Total Patients - number of patients in the data
- Total Admissions - number of admission records
- Average Billing - average billing amount
- Average Age - average age of patients
- Average Stay - average number of days a patient stayed

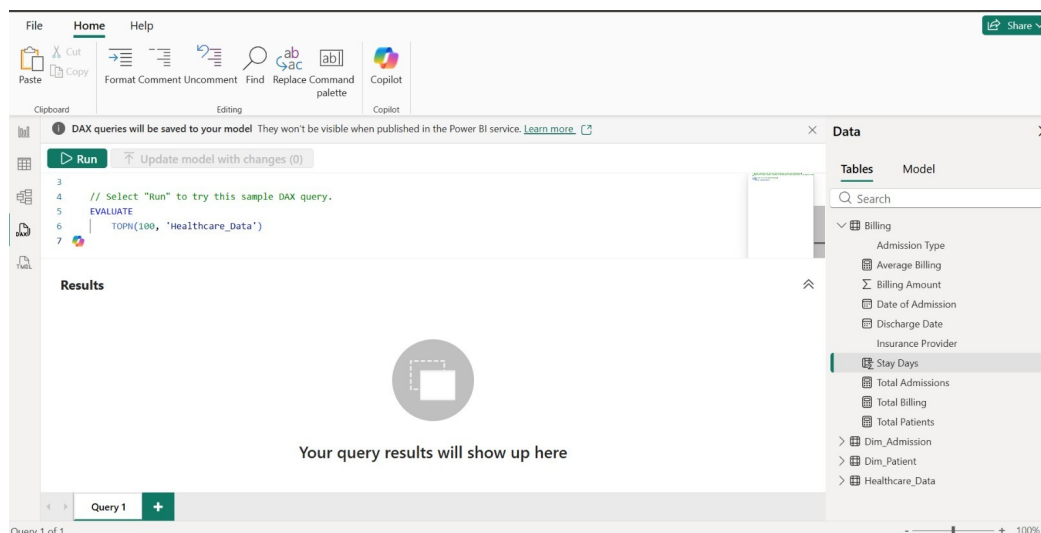


Figure 3: DAX query view and the measures in the Billing table

4. Building the Dashboard

I built the dashboard step by step. First I added the KPI cards at the top so the main numbers can be seen at once. After that I added the slicers and then the charts below them. Later I adjusted the size and spacing of the visuals to make the page look cleaner.

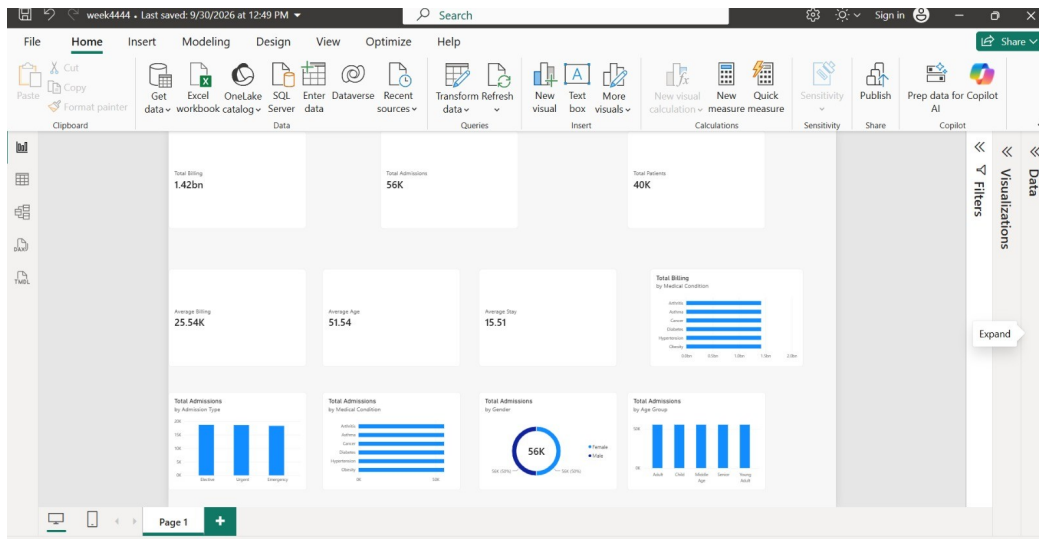


Figure 4: Dashboard in the middle of development, with KPI cards and charts

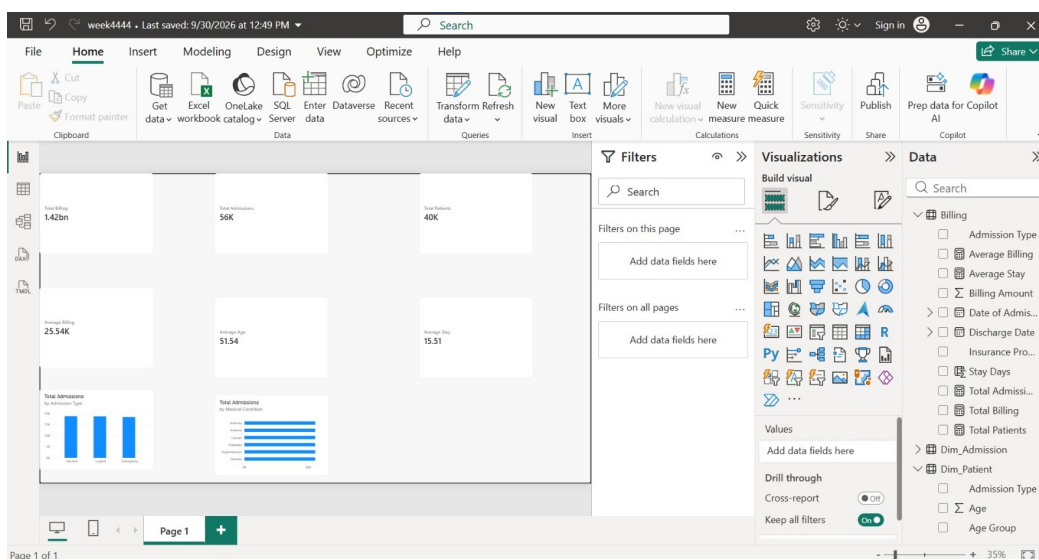


Figure 5: Layout after adjusting the spacing

5. Final Dashboard

The final dashboard has the KPI cards, slicers and charts together on one page. The slicers are for admission date, medical condition, gender and admission type. The charts show billing by medical condition, admissions by admission type, admissions by medical condition, admissions by gender and admissions by age group.

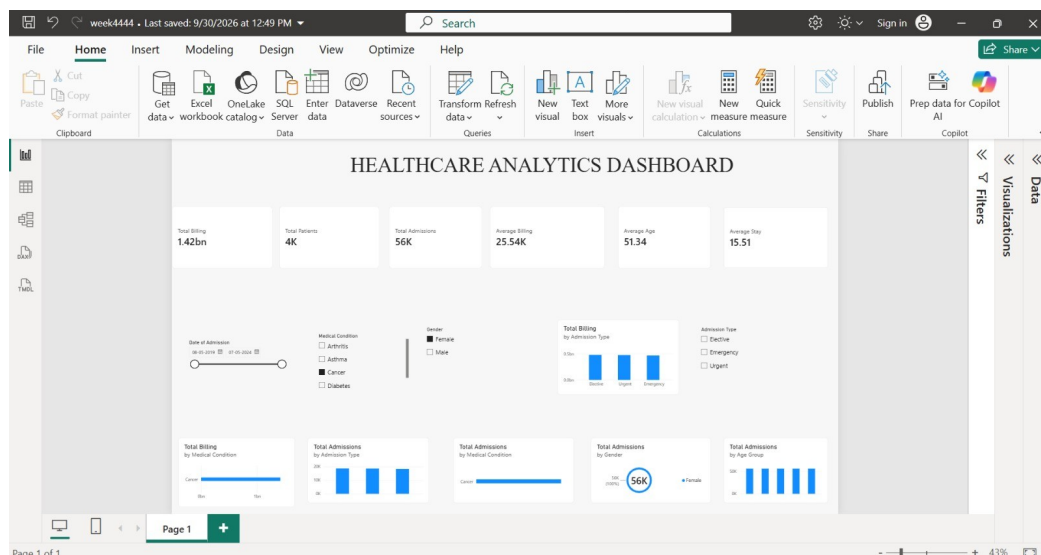


Figure 6: Final dashboard

What the page contains:

- KPI cards for Total Billing, Total Patients, Total Admissions, Average Billing, Average Age and Average Stay
- A date slicer to pick the admission date range
- Slicers for medical condition, gender and admission type
- A chart of billing by medical condition
- Charts of admissions by admission type and by medical condition
- Gender and age group charts for the demographics

6. Testing

After adding the measures and charts I checked the dashboard to make sure it worked properly. I looked at the model view to confirm all the tables and relationships were there, and then clicked through the slicers to see that the visuals changed with each selection. I also checked that the KPI cards were showing values and that the whole dashboard fits on a single page.

7. Conclusion

In Week 5 I converted the healthcare data into an interactive Power BI dashboard. The work covered the data model, DAX measures, KPI cards, slicers and several charts. The final page gives an overall view of the patient, admission and billing data.

For Week 6 I will continue in the same Power BI file. I will reuse the tables, relationships, measures and this dashboard page, and add the new Week 6 requirements on top of it.